

Dr. Babasaheb Ambedkar Technological University, Lonere

Electrical Engineering Department

Summer Examination May-2015

Subject: High Voltage Engineering [Old]

Date: 16/5/2015

SEM: VIII

Max. Marks: 50

Instructions:

- 1.) Attempt any one question from (Q.1 and Q.2), (Q.3 and Q.4), (Q.5 and Q.6), (Q.7 and Q.8)
- 2.) Attempt any one question from the remaining.
- 3.) Assume suitable data wherever necessary and mention it clearly.

- Q.1) a.)** Explain the different secondary ionization processes occurs in gaseous insulating medium during its breakdown. (05)
- b.)** Discuss the Townsend's criterion for breakdown in gaseous medium. (05)
- Q.2)** Explain the concept of pure and commercial liquid and give the process of purification of commercial liquid describe conduction and breakdown in pure liquids. (10)
- Q.3)** Discuss the various breakdown mechanism due to which in solid insulating medium fails. (10)
- Q.4)** Explain the operational principle of Van de Graaff generator along with its constructional details for generation high D.C. voltages. (10)
- Q.5)** Explain the Hall generator for measuring high d.c. currents with principle of operation and construction along with neat diagram. (10)
- Q.6)** Write a note on the attenuation and distortion of travelling wave in high voltage engineering. (10)
- Q.7)** Explain the different test performed on bushing in a high voltage-engineering laboratory. (10)
- Q.8)** A transmission line has the following line constants $R = 0.1 \text{ ohm/km}$, $L = 1.26 \text{ mH/km}$, $C = 0.009 \text{ } \mu\text{F/km}$, and $G = 0$. If the line is a 3-phase line and is charged from one end at a line voltage of **230 kV**, find the rise in voltage at the other end, if the line length is **400 km**. [10]

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